

L8

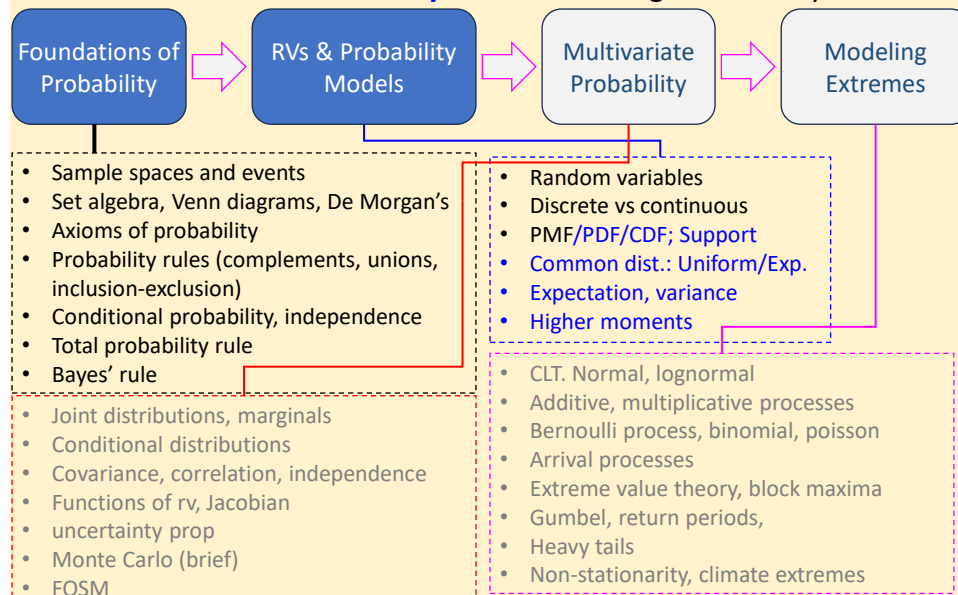
Aug 11, 2026

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Module I: Chapter 2 (M1C2)

Module 1: **Probability** – Understanding Uncertainty



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Messages from the Last Class

• **Message 1:** *What's in a name?*

What's wrong with the term "random variable"?

- They are neither **random** nor **variables**
- On the contrary, they are **well-defined** and **functions**
- Then why call them random variables?
 - They are **probabilistic counterpart** of **deterministic variables** which take a *fixed value once determined*.
 - RV's instead start with an experiment and map each event to a real number.
 - RV contains **information on how probable are different events**.

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Bernoulli rv

→ Takes only two values

0, 1

e.g. coin toss: Head as 1

fail/pass:

→ Single parameter RV: p

→ PMF of a Bernoulli rv is

$$p_X(x) = \begin{cases} p, & x=1 \\ 1-p, & x=0 \end{cases}$$

for a fair coin, $p = 1/2$

for a biased/loading coin, $p \leq 1/2$

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Binomial rv

- Toss a coin n times;
- Each coin toss results in **success with probability p**
 - Each coin toss is a Bernoulli trial
- Each **toss is independent** of each other
- X is the number of successes in n independent & identical trials; then X is a **Binomial rv** with
 - Two parameters: n, p
- PMF is

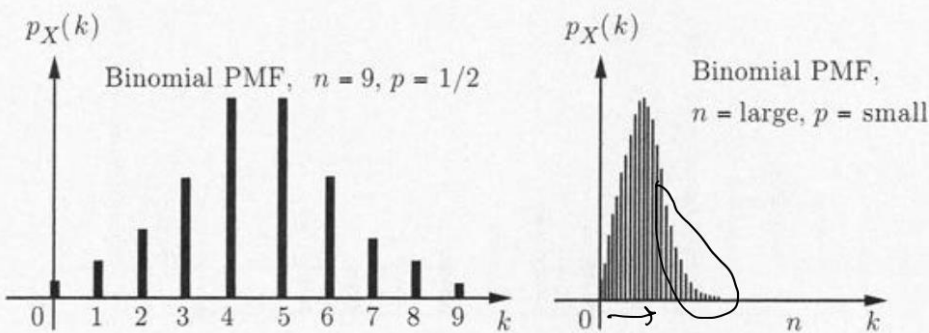
$$p_X(k) = \Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n$$
 - Important to write the range
- Show that $\sum_x p_X(x) = 1$

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Binomial rv

- $p = \frac{1}{2}$, the PMF is symmetric around $\frac{n}{2}$.
- If $p < \frac{1}{2}$, the PMF is skewed toward 0.
- If $p > \frac{1}{2}$, the PMF is skewed toward n .



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Geometric rv

- X is the **number of tosses needed for the first head**
- X is the number of trials needed for the first success
identical & independent trials; then X is a Geometric rv

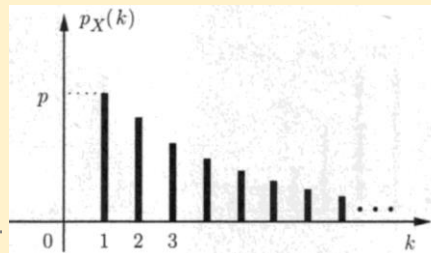
- PMF is

$$p_X(k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

- Important to write the range

- Show that $\sum_k p_X(k) = 1$

$$\sum_{k=1}^{\infty} p_X(k) = \sum_{k=1}^{\infty} (1 - p)^{k-1}p = 1$$



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Poisson rv

- Given λ : average number of successes in a time window
- X is the rv for the number of successes

- The PMF is

$$p_X(k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots \text{ \& } \lambda > 0$$

- Is this a legitimate PMF?

- Show that $\sum_k p_X(k) = 1$

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